WEIGHT-SPACE VIEW

- Until now we considered a hypothesis space $\mathcal{H}$ of parameterized functions $f(\mathbf{x} \mid \boldsymbol{\theta})$ (in particular, the space of linear functions).
- Using Bayesian inference, we derived distributions for $\boldsymbol{\theta}$ after having observed data $\mathcal{D}$.
- Prior believes about the parameter are expressed via a prior distribution $q(\boldsymbol{\theta})$, which is updated according to Bayes' rule

$$\underbrace{p(\boldsymbol{\theta}|\mathbf{X},\mathbf{y})}_{\text{posterior}} = \frac{\overbrace{p(\mathbf{y}|\mathbf{X},\boldsymbol{\theta})}^{\text{likelihood}}\,\overbrace{q(\boldsymbol{\theta})}^{\text{prior}}}{\underbrace{p(\mathbf{y}|\mathbf{X})}_{\text{marginal}}}.$$

FUNCTION-SPACE VIEW

Let us change our point of view:

- Instead of "searching" for a parameter $\boldsymbol{\theta}$ in the parameter space, we directly search in a space of "allowed" functions $\mathcal{H}$.
- We still use Bayesian inference, but instead specifying a prior distribution over a parameter, we specify a prior distribution **over functions** and update it according to the data points we have observed.

WEIGHT-SPACE VS. FUNCTION-SPACE VIEW

Intuitively, imagine we could draw a huge number of functions from some prior distribution over functions (*).

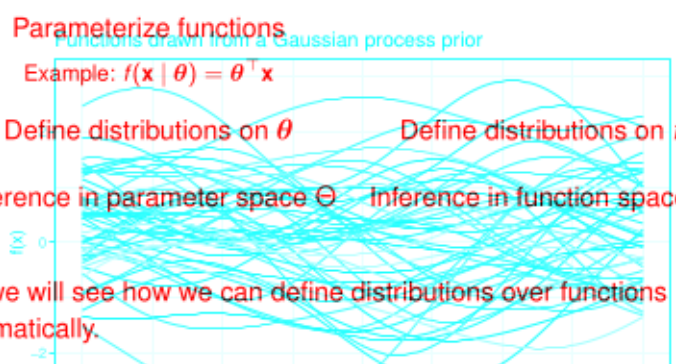
Weight-Space View **Function-Space View**

Parameterize functions

Example: $f(\mathbf{x} \mid \boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{x}$

Define distributions on $\boldsymbol{\theta}$ Define distributions on f

Inference in parameter space Θ Inference in function space $\mathcal{H}$

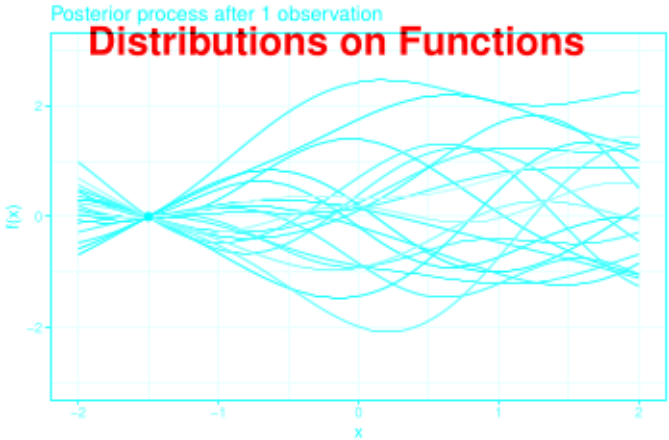


Next, we will see how we can define distributions over functions mathematically.

(*) We will see in a minute how distributions over functions can be specified.

FUNCTION-SPACE VIEW

After observing some data points, we are only allowed to sample those functions, that are consistent with the data.



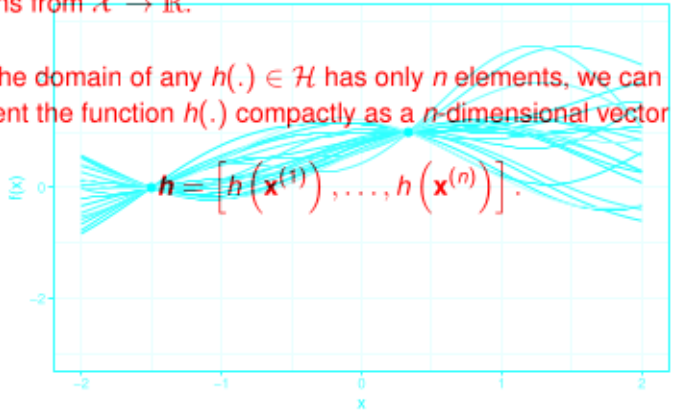
DISCRETE FUNCTIONS

For simplicity, let us consider functions with finite domains first.

Let $\mathcal{X} = \{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\}$ be a finite set of elements and $\mathcal{H}$ the set of all functions from $\mathcal{X} \to \mathbb{R}$.

Since the domain of any $h(.) \in \mathcal{H}$ has only n elements, we can represent the function $h(.)$ compactly as a n-dimensional vector

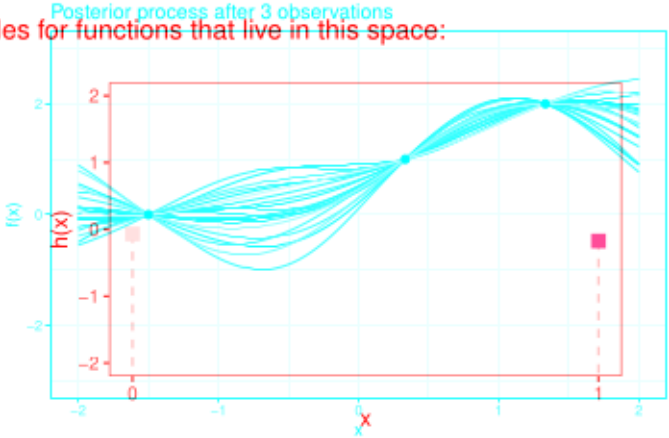
$$\boldsymbol{h} = \left[h\left(\mathbf{x}^{(1)}\right), \ldots, h\left(\mathbf{x}^{(n)}\right)\right].$$



DISCRETE FUNCTIONS

Example 1: Let us consider $h : \mathcal{X} \to \mathcal{Y}$ where the input space consists of **two** points $\mathcal{X} = \{0, 1\}$.

Examples for functions that live in this space:



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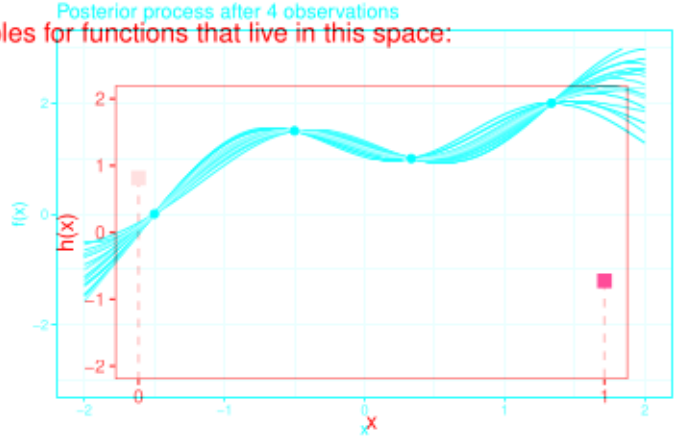
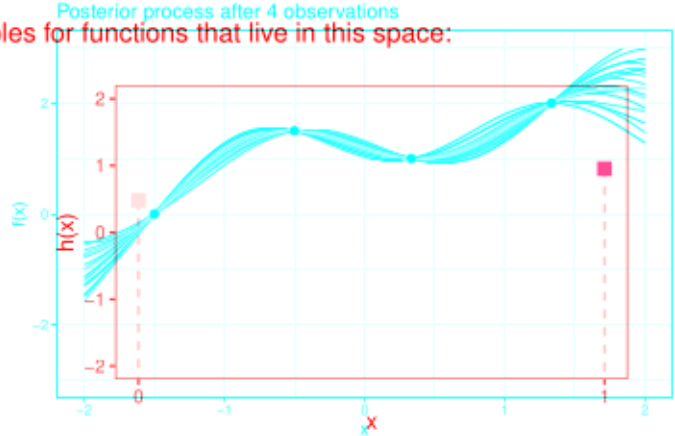


DISCRETE FUNCTIONS

Example 1: Let us consider $h \sim \mathcal{N}(\mu, \Sigma)$ where the input space consists of **two** points $\mathcal{X} = \{0, 1\}$.

Examples for functions that live in this space:



DISCRETE FUNCTIONS

Example 2: Let us consider $h : \mathcal{X} \to \mathcal{Y}$ where the input space consists of **five** points $\mathcal{X} = \{0, 0.25, 0.5, 0.75, 1\}$.

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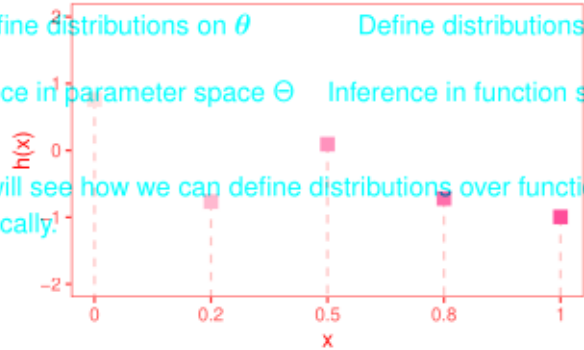

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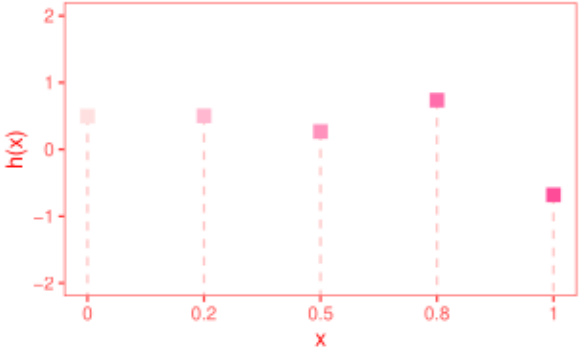
Distributions on Functions



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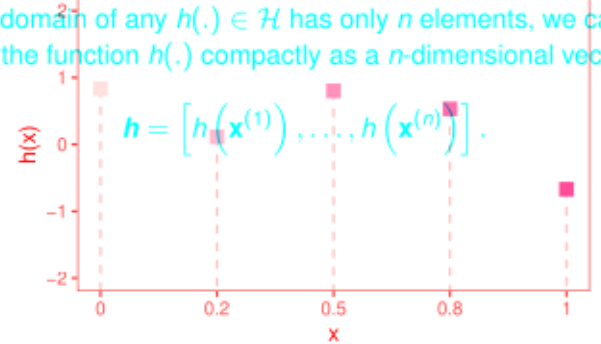

DISCRETE FUNCTIONS

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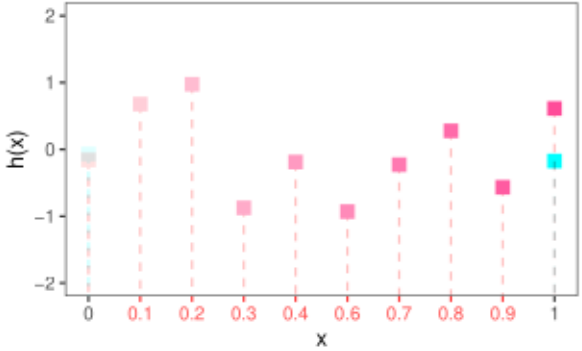

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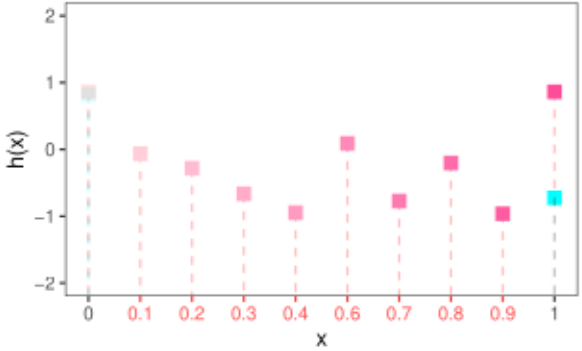

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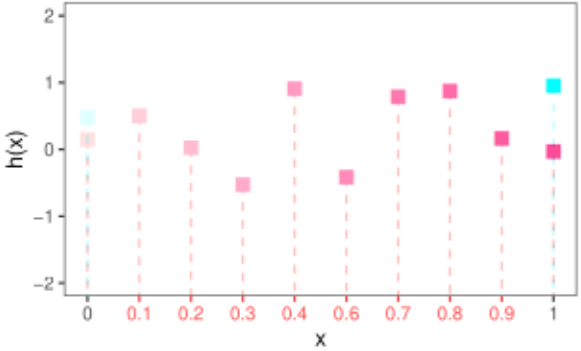

DISTRIBUTIONS ON DISCRETE FUNCTIONS

Example 2: Let us consider $h : \mathcal{X} \to \mathcal{Y}$ where the input space consists of 5 points $\mathcal{X} = \{0, 0.25, 0.5, 0.75, 1\}$.

One natural way to specify a probability function on discrete function $h \in \mathcal{H}$ is to use the vector representation

Examples for functions that live in this space:

$$\boldsymbol{h} = \left[h\left(\mathbf{x}^{(1)}\right), h\left(\mathbf{x}^{(2)}\right), \ldots, h\left(\mathbf{x}^{(n)}\right)\right]$$

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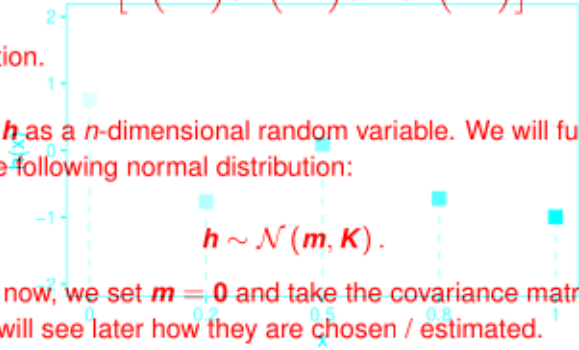


Let us see $\boldsymbol{h}$ as a n-dimensional random variable. We will further assume the following normal distribution:

$$\boldsymbol{h} \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K}).$$

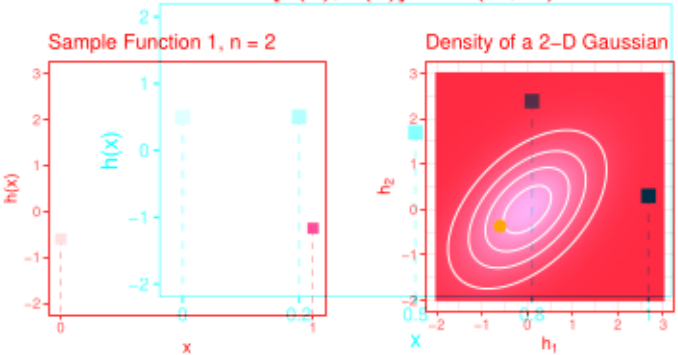
Note: For now, we set $\boldsymbol{m} = \mathbf{0}$ and take the covariance matrix $\boldsymbol{K}$ as given. We will see later how they are chosen / estimated.

DISCRETE FUNCTIONS

Example 1 (continued): Let $h : \mathcal{X} \to \mathcal{Y}$ be a function that is defined on **two** points $\mathcal{X}$. We sample functions by sampling from a two-dimensional normal variable

Examples for functions that live in this space:

$$\boldsymbol{h} = [h(1), h(2)] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K})$$



In this example, $m = (0, 0)$ and $K = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$.

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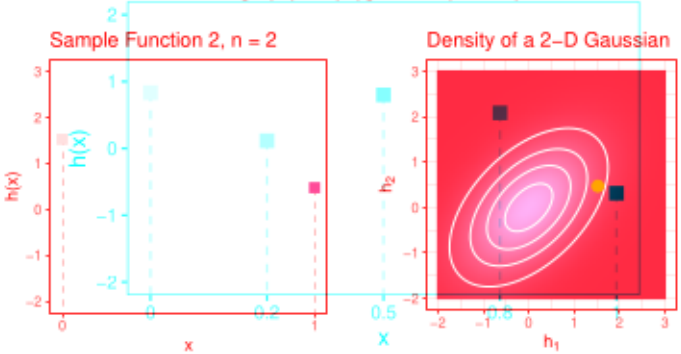

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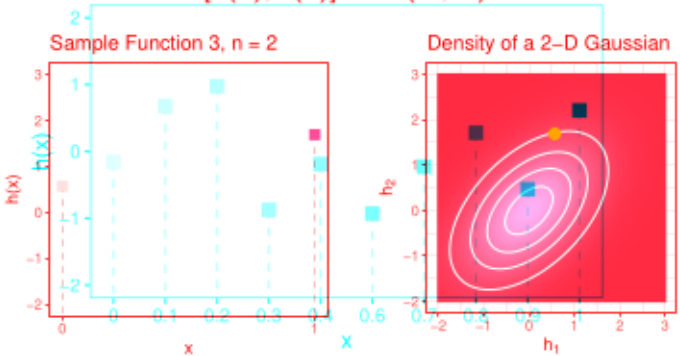

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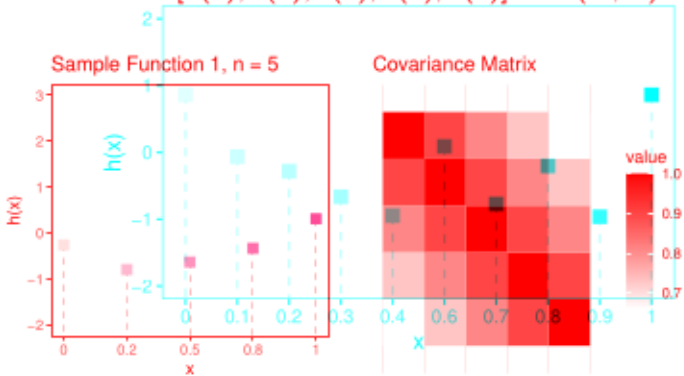

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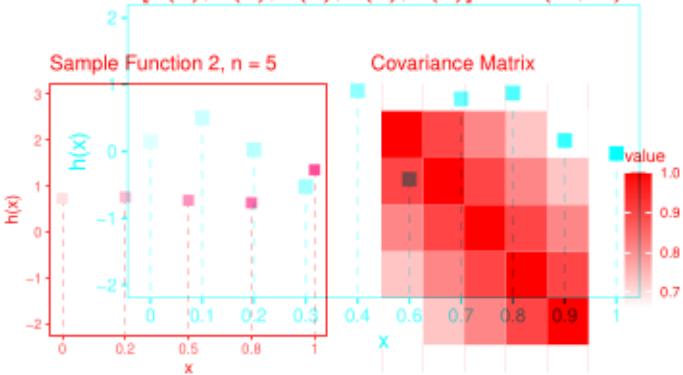

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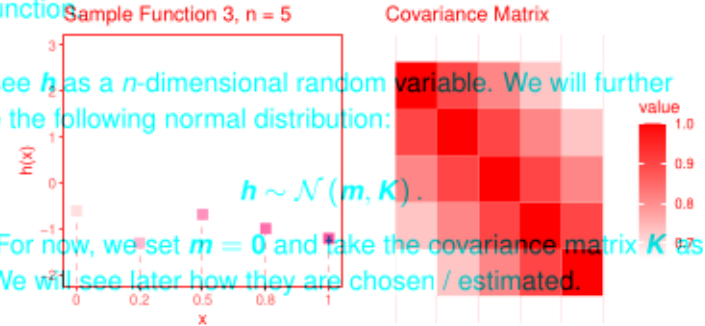

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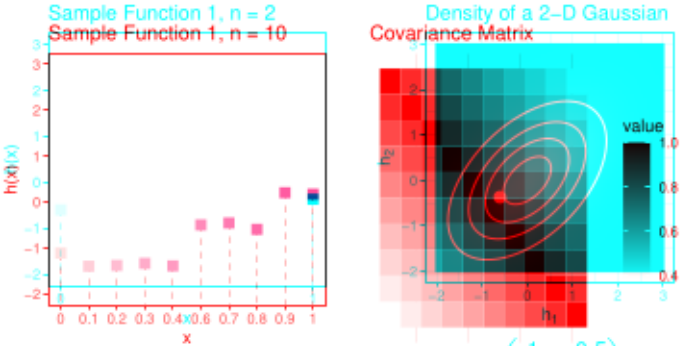

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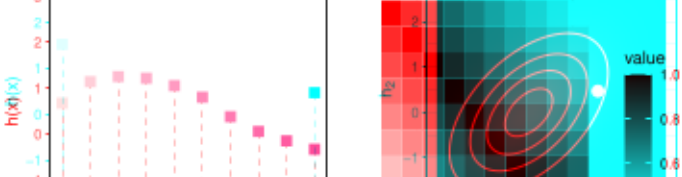

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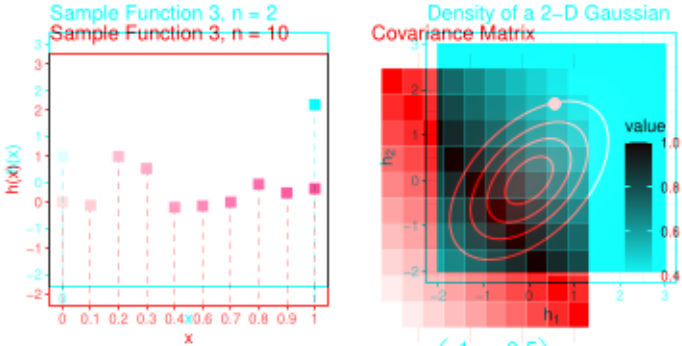

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ROLE OF THE COVARIANCE FUNCTION

Note that the covariance controls the "shape" of the drawn function.
Consider two extreme cases where function values are

a) strongly correlated: $\boldsymbol{K} = \begin{pmatrix} 1 & 0.99 & \dots & 0.99 \\ 0.99 & 1 & \dots & 0.99 \\ 0.99 & 0.99 & \dots & 0.99 \\ 0.99 & \dots & 0.99 & 1 \end{pmatrix}$

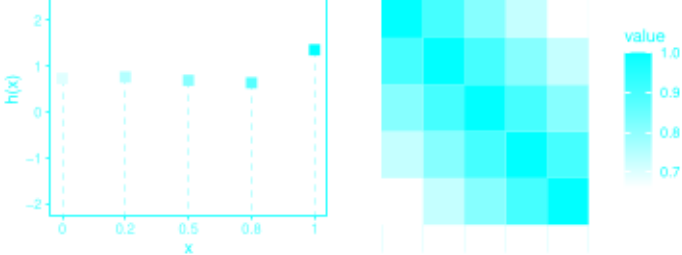
b) uncorrelated: $\boldsymbol{K} = \boldsymbol{I}$

DISCRETE FUNCTIONS

Example 2 (continued): Let us consider $h : \mathcal{X} \to \mathcal{Y}$ where the input space consists of **five** points. We sample functions by sampling from a five-dimensional normal variable

Gaussian Processes

$$\boldsymbol{h} = [h(1), h(2), h(3), h(4), h(5)] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K})$$



FROM DISCRETE TO CONTINUOUS FUNCTIONS

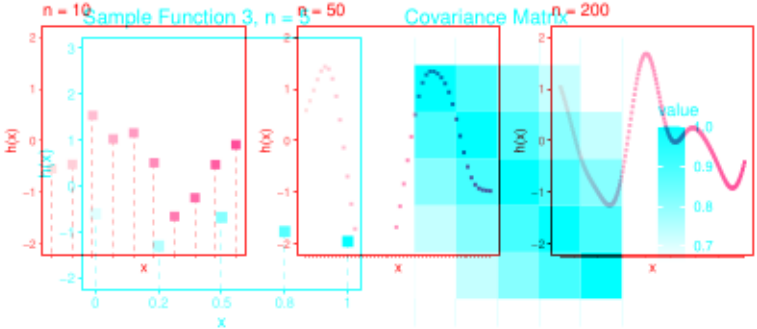
Example 2 (continued): Let us consider $h : \mathcal{X} \to \mathbb{R}$ where the input space consists of five points. We sample functions by sampling from a five-dimensional normal variable

- We defined distributions on functions with discrete domain by defining a Gaussian on the vector of the respective function values

$$\mathbf{h} = [h(\mathbf{x}^{(1)}), h(\mathbf{x}^{(2)}), \ldots, h(\mathbf{x}^{(n)})] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K})$$

$$\boldsymbol{h} = [h(1), h(2), h(3), h(4), h(5)] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K})$$

- We can do this for $n \to \infty$ (as "granular" as we want)

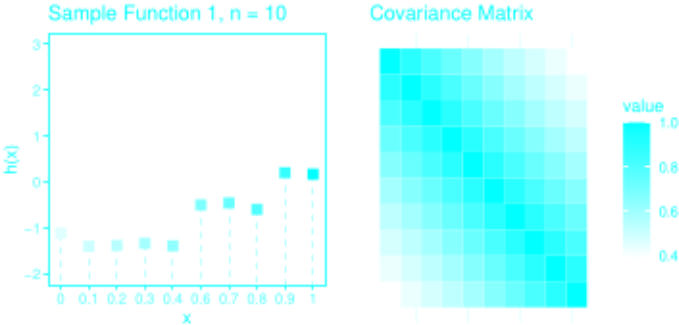


FROM DISCRETE TO CONTINUOUS FUNCTIONS

Example 3 (continued): Let us consider $h : \mathcal{X} \to \mathcal{Y}$ where the input space consists of ten points. We sample functions by sampling from ten-dimensional normal variable.

- No matter how large n is, we are still considering a function over a discrete domain.
- How can we extend our definition to functions with **continuous domain** $\mathcal{X} \subset \mathbb{R}$?

$$\boldsymbol{h} = [h(1), h(2), \ldots, h(10)] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K})$$

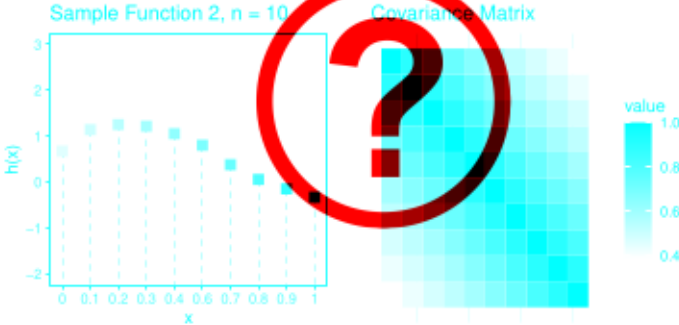


GAUSSIAN PROCESSES: INTUITION

Example 3 (continued): Let us consider $h : \mathcal{X} \to \mathbb{R}$ where the input space consists of ten points. We sample functions by sampling from ten-dimensional normal variable

- Intuitively, a function f drawn from **Gaussian process** can be understood as an "infinite" long Gaussian random vector.
- It is unclear how to handle an "infinite" long Gaussian random vector!

$$\boldsymbol{h} = [h(1), h(2), \ldots, h(10)] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K})$$



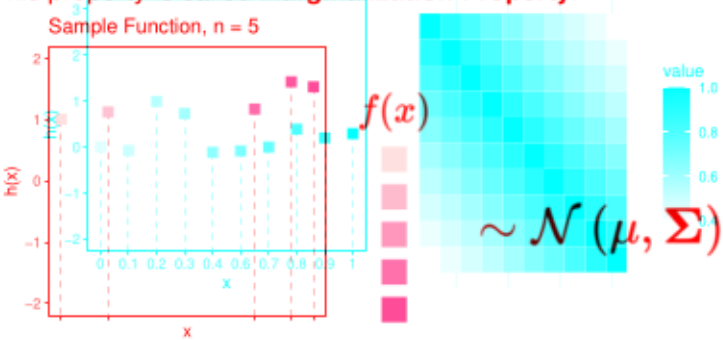
GAUSSIAN PROCESSES: INTUITION

- Thus, it is required that for **any finite set** of inputs $\left(\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right) \in \mathcal{X}^n$, the vector $\boldsymbol{f}$ has a Gaussian distribution

$$\boldsymbol{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K}),$$

with $\boldsymbol{m}$ and $\boldsymbol{K}$ being calculated by a mean function $m(.)$ / covariance function $k(.,.)$.

- This property is called **Marginalization Property**.



GAUSSIAN PROCESSES: INTUITION

Note that the required property holds for any finite set of inputs.

- Thus, it is required that for **any finite set** of inputs $[\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}] = X$, the vector $\boldsymbol{f}$ has a Gaussian distribution

$$\boldsymbol{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K}),$$

Consider two extreme cases of the covariance function $\boldsymbol{K}$:

a) strongly correlated: $\boldsymbol{K} = \begin{pmatrix} 1 & 0.99 & \cdots & 0.99 \\ 0.99 & 1 & \cdots & 0.99 \\ 0.99 & 0.99 & \cdots & 0.99 \\ 0.99 & \cdots & 0.99 & 1 \end{pmatrix}$

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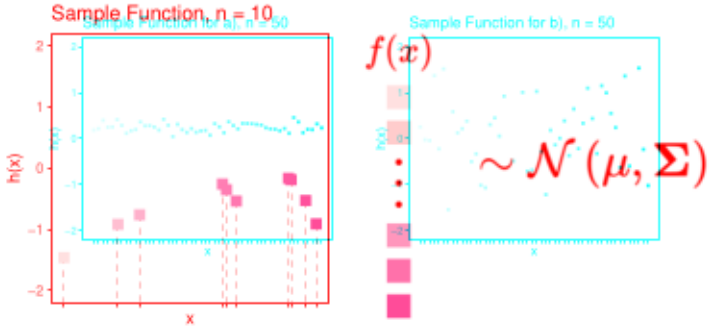


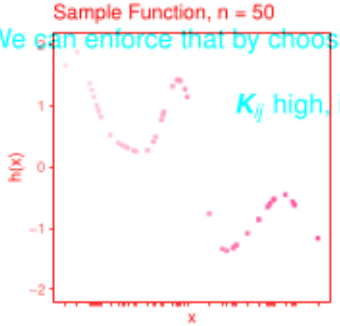
GAUSSIAN PROCESSES: INTUITION

- Thus, it is required that for **any finite set** of inputs $\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\} \subset \mathcal{X}$, the vector f has a Gaussian distribution

  $$f = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K}),$$

  with $\boldsymbol{m}$ and $\boldsymbol{K}$ being calculated by a mean function $m(.)$ / covariance function $k(.,.)$.
- This property is called **Marginalization Property**.
- We can enforce that by choosing a covariance function with

  $\boldsymbol{K}_{ij}$ high, if $\mathbf{x}^{(i)}, \mathbf{x}^{(j)}$ close.



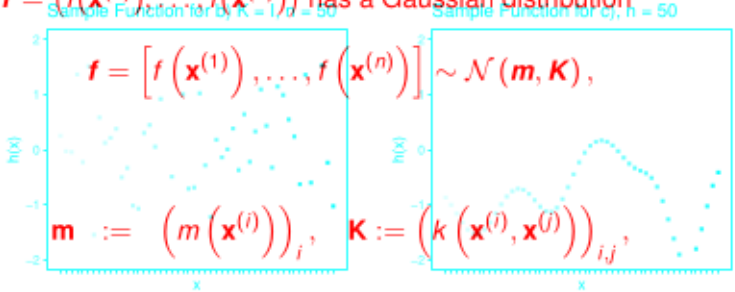
$$f(x)$$

$$\sim \mathcal{N}(\mu, \Sigma)$$

GAUSSIAN PROCESSES

This intuitive explanation is formally defined as follows:

A function $f(\mathbf{x})$ is generated by a GP $\mathcal{GP}\left(m(\mathbf{x}), k\left(\mathbf{x}, \mathbf{x}'\right)\right)$ if for **any finite** set of inputs $\left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\}$, the associated vector of function values $\boldsymbol{f} = \left(f(\mathbf{x}^{(1)}), \ldots, f(\mathbf{x}^{(n)})\right)$ has a Gaussian distribution



$$\boldsymbol{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K}),$$

with

$$\mathbf{m} := \left(m\left(\mathbf{x}^{(i)}\right)\right)_i, \quad \mathbf{K} := \left(k\left(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}\right)\right)_{i,j},$$

where $m(\mathbf{x})$ is called mean function and $k(\mathbf{x}, \mathbf{x}')$ is called covariance function.

Note: $k(\cdot, \cdot)$ is known as the **covariance function** or **kernel**. It will be studied in more detail later on.

SAMPLING FROM A GAUSSIAN PROCESS PRIOR

We can draw functions from a Gaussian process prior. Let us consider $f(\mathbf{x}) \sim \mathcal{GP}\left(0, k(\mathbf{x}, \mathbf{x}')\right)$ with the squared exponential covariance function [*]

Gaussian Processes

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{1}{2\ell^2}\|\mathbf{x} - \mathbf{x}'\|^2\right), \quad \ell = 1.$$

This specifies the Gaussian process completely.

[*] We will talk later about different choices of covariance functions.

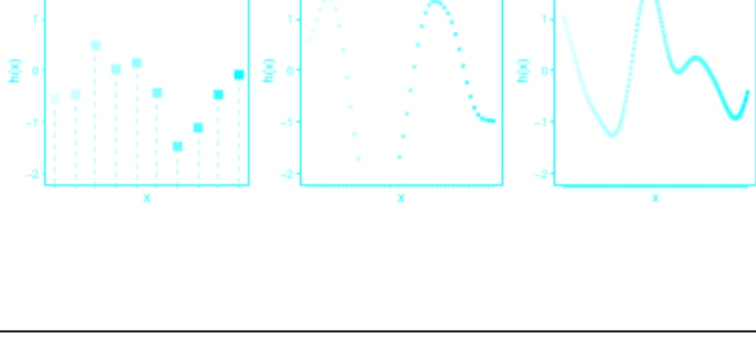
FROM DISCRETE TO CONTINUOUS FUNCTIONS

- We defined distributions on functions with discrete domain by defining a Gaussian on the vector of the respective function values

$$\mathbf{h} := [h(\mathbf{x}^{(1)}), h(\mathbf{x}^{(2)}), \dots, h(\mathbf{x}^{(n)})] \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

Gaussian Processes as Indexed Family

- We can do this for $n \to \infty$ (as "granular" as we want)



GAUSSIAN PROCESSES AS AN INDEXED FAMILY

- No matter how large n is, we are still considering a function over a discrete domain.
- How can we extend our definition to functions with **continuous domain** $\mathcal{X} \subset \mathbb{R}$?

A Gaussian process is a special case of a **stochastic process** which is defined as a collection of random variables indexed by some index set (also called an **indexed family**). What does it mean?

An **indexed family** is a mathematical function (or "rule") to map indices $t \in T$ to objects in $\mathcal{S}$.

Definition

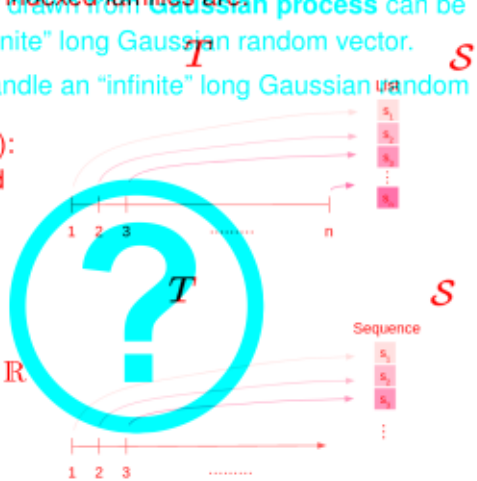
A **family of elements in $\mathcal{S}$ indexed by T** (indexed family) is a surjective function

$$\begin{aligned} s : T &\rightarrow \mathcal{S} \\ t &\mapsto s_t = s(t) \end{aligned}$$

INDEXED FAMILY PROCESSES: INTUITION

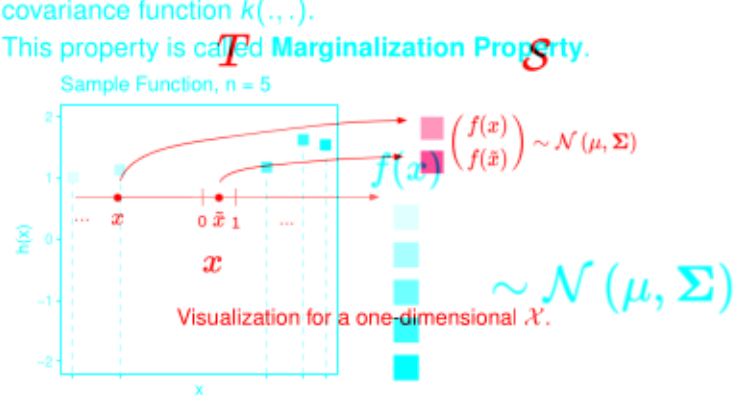
Some simple examples for indexed families are:

- Intuitively, a function f drawn from **Gaussian process** can be understood as an "infinite" long Gaussian random vector.
- It is unclear how to handle an "infinite" long Gaussian random vector!
- finite sequences (lists): $T = \{1, 2, \ldots, n\}$ and $(s_t)_{t \in T} \in \mathbb{R}$
- infinite sequences: $T = \mathbb{N}$ and $(s_t)_{t \in T} \in \mathbb{R}$



INDEXED FAMILY PROCESSES: INTUITION

- Thus, it is required that for any finite set of inputs $\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\} \subset \mathcal{X}$, the vector follows a Gaussian distribution $\mathbf{f} = [f(\mathbf{x}^{(1)}), \ldots, f(\mathbf{x}^{(n)})]^T \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$, with covariance matrix $\mathbf{K} = [k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})]_{ij}$ generated by a mean function $m(.)$ / covariance function $k(.,.)$.
- This property is called **Marginalization Property**.



Visualization for a one-dimensional $\mathcal{X}$.

INDEXED FAMILY

- A Gaussian process is also an indexed family, where the random variables $f(\mathbf{x})$ are indexed by the input values $\mathbf{x} \in \mathcal{X}$.
- Their special feature: Any indexed (finite) random vector has a multivariate Gaussian distribution (which comes with all the nice properties of Gaussianity!).
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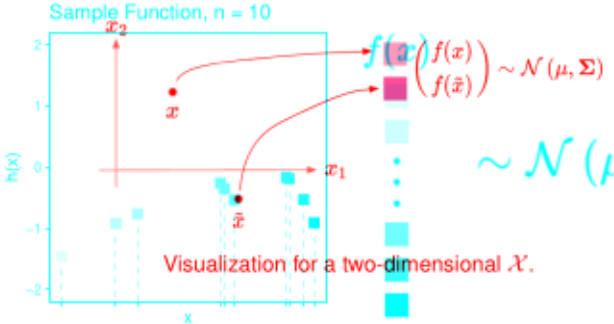

$$f(x) \begin{pmatrix} f(x) \\ f(\tilde{x}) \end{pmatrix} \sim \mathcal{N}(\mu, \mathbf{\Sigma})$$

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Visualization for a two-dimensional $\mathcal{X}$.

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- Thus, it is required that for **any finite set** of inputs $\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\} \subset \mathcal{X}$, the vector $\mathbf{f}$ has a Gaussian distribution

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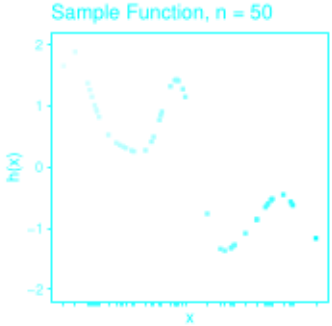

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where $m(\mathbf{x})$ is called mean function and $k(\mathbf{x}, \mathbf{x}')$ is called covariance function.

GAUSSIAN PROCESSES

A GP is thus **completely specified** by its mean and covariance function

$$
\begin{aligned}
m(\mathbf{x}) &= \mathbb{E}[f(\mathbf{x})] \\
k(\mathbf{x}, \mathbf{x}') &= \mathbb{E}\Big[\big(f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})]\big)\big(f(\mathbf{x}') - \mathbb{E}[f(\mathbf{x}')]\big)\Big]
\end{aligned}
$$

Note: For now, we assume $m(\mathbf{x}) \equiv 0$. This is not necessarily a drastic limitation - thus it is common to consider GPs with a zero mean function.

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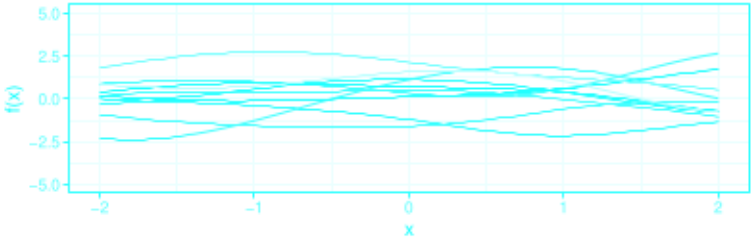
SAMPLING FROM A GAUSSIAN PROCESS PRIOR

To visualize a sample function, we

- choose a high number n (equidistant) points $\left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\}$
- compute the corresponding covariance matrix $\mathbf{K} = \left(k\left(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}\right)\right)_{i,j}$ by plugging in all pairs $\mathbf{x}^{(i)}, \mathbf{x}^{(j)}$
- sample from a Gaussian $\boldsymbol{f} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{K})$.

We draw 10 times from the Gaussian, to get 10 different samples.



Since we specified the mean function to be zero $m(\mathbf{x}) \equiv 0$, the drawn functions have zero mean.

Gaussian Processes as Indexed Family

GAUSSIAN PROCESSES AS AN INDEXED FAMILY

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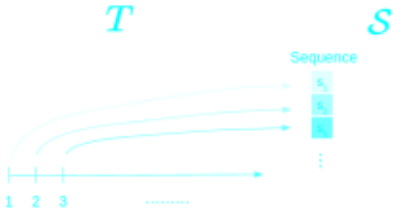
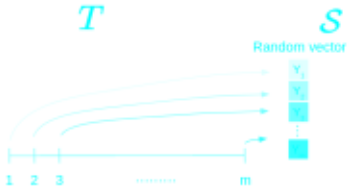


INDEXED FAMILY

But the indexed set $\mathcal{S}$ can be something more complicated, for example functions or **random variables** (RV):

- $T = \{1, \ldots, m\}$, Y_t's are RVs: Indexed family is a random vector.
- $T = \{1, \ldots, m\}$, Y_t's are RVs: Indexed family is a stochastic process in discrete time
- $T = \mathbb{Z}^2$, Y_t's are RVs: Indexed family is a 2D-random walk.



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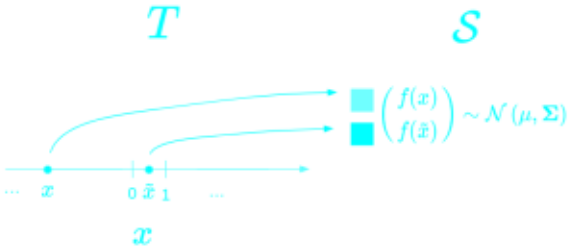

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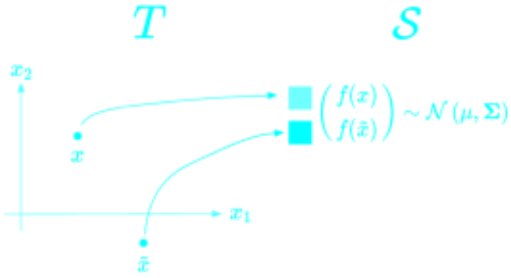

Visualization for a two-dimensional $\mathcal{X}$.